

Reflexive

R is reflexive, iff $(a, a) \in R$ for each $a \in A$

②

Symmetric

R is sym iff $(a, b) \in R \rightarrow (b, a) \in R$ for each $a, b \in A$

Transitive

R is trans iff $(a, b) \in R \wedge (b, c) \in R \rightarrow (a, c) \in R$ for all $a, b, c \in A$

Antisymmetric

R is anti

iff

$(a, b) \in R \wedge (b, a) \in R \rightarrow a = b$ for all $a, b \in A$



Partial Order PO reflexive, transitive, antisymmetric

Equivalence Relation EQ reflexive, symmetric, transitive

7 7

(3)

$$(- * -) + (- * -) + (- * -) \leftarrow \{0, 1\}$$

01	*	min
0*0	0	0
0*1	0	0
1*0	0	0
1*1	1	1
	+	max
00	0	0
01	1	1
10	1	1
11	(2)	1

$$\{1, 3, 5, 7\} \leq$$

$$\{1, 3, 5, 7\} \geq$$

$$\{1, 3, 5, 7\} =$$

Max

7

1

$\{1, 3, 5, 7\}$

forall a

forall a

$7 \notin a$

$1 \notin a$

Min ④

1

7

$\{1, 3, 5, 7\}$

(5)

$$2_{10} + 3_{10} = 10_2 + 10_2 = 101_2 = 5_{10}$$

$$-2_{10} + 3_{10} = 1110_2 + 11_2 = \underline{0001}_2 = 1_{10}$$

$$-2_{10} + -3_{10} = 1110_2 + 1101_2 = \underline{0101}_2 = -5_{10}$$

$$= 0_{16} =$$

$$\boxed{8 - 8}_{16}$$

$$\frac{\boxed{10000}_2}{\boxed{1111}_2} = 1$$